

The Chain Rule (Multivariable Version)

Benjamin Abshire[†]

March 6, 2026

I Introduction

If $f(x_1, x_2, \dots, x_n)$ then

$$\frac{\partial f}{\partial x_j} = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_j, x_{j+h}, \dots, x_n) - f(x_1, \dots, x_j, x_n)}{h} \quad (1)$$

We call this a partial derivative of f with respect to x_j . We regard all variables besides the ones we are differentiating with respect to as constants. For example, if we are differentiating $f(x, y)$ with respect to x , we consider y a constant. If we have a multi-variable composite function, then we can use the chain rule to find the derivative with respect to whichever variable we want.

If we let $z = f(x_1, \dots, x_n)$ with $x_j = x_j(t)$ (i.e. x is a function of t), then the chain rule gives

$$\frac{dz}{dt} = \left(\frac{\partial z}{\partial x_1} \right) \frac{dx_1}{dt} + \left(\frac{\partial z}{\partial x_n} \right) \frac{dx_n}{dt} \quad (2)$$

The main goal of the multivariable chain rule is that if we want to see how z changes with x_n but x_n depends on t we must also account for the change in x_n when we change t .

II Examples

Example 1. Let $z = x^2 + 2xy$, $x = \sin t$, $y = 2e^t$. Find $\frac{dz}{dt}$.

Solution. Applying Equation 2, we get

$$\frac{dz}{dt} = \left(\frac{\partial z}{\partial x} \right) \frac{dx}{dt} + \left(\frac{\partial z}{\partial y} \right) \frac{dy}{dt} \implies \frac{dz}{dt} = (2x + 2y)(\cos t) + (2x)(e^t)$$

Note that we keep the derivative in terms of x, y, t for simplicity.

[†]Definitions and example problems from MATH 22005: Analytic Geometry and Calculus III instructed by Dr. Gang Yu. Kent State University.

Example 2. Let $w = xe^{y^2z}$, $x = t^2$, $y = 2t$, $z = 3 - t$. Find $\frac{dw}{dt}$.

Solution. Now that we've added another variable, the process does not change, we just have to do an extra step. Applying Equation 2, we get

$$\begin{aligned}\frac{dw}{dt} &= \left(\frac{\partial w}{\partial x}\right)\frac{dx}{dt} + \left(\frac{\partial w}{\partial y}\right)\frac{dy}{dt} + \left(\frac{\partial w}{\partial z}\right)\frac{dz}{dt} \\ \implies & (e^{y^2z})(2t) + (xe^{y^2z} \cdot 2yz)(2) + (xe^{y^2z} \cdot y^2)(-1)\end{aligned}$$

What if $f(x, y) = z$, $x = x(s, t)$, $y = (s, t)$? In this case (where both x and y are functions of two variables) we use ONLY partial derivatives, and must now find the derivatives with respect to s and t . These formulas are

$$\begin{aligned}\frac{\partial z}{\partial t} &= \left(\frac{\partial z}{\partial x}\right)\frac{\partial x}{\partial t} + \left(\frac{\partial z}{\partial y}\right)\frac{\partial y}{\partial t} \\ \frac{\partial z}{\partial s} &= \left(\frac{\partial z}{\partial x}\right)\frac{\partial x}{\partial s} + \left(\frac{\partial z}{\partial y}\right)\frac{\partial y}{\partial s}\end{aligned}$$

Note that this is the same formula as example 1, but just with all partial derivatives with respect to if we are finding t or s . We can add in as many variables as we want into x and y , and for each one we just repeat the formula but with respect to the new variable.